Joseph Wu

Senior Software Engineer | AI Engineering

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English and Mandarin

PROFILE

Senior software engineer with fourteen years building backend platforms, distributed services, and AI-enabled applications. Python runs through the work, from API design and data processing to cloud deployment and production operations, alongside Java and Node.js services. Recent work covers retrieval-augmented generation, LLM integrations, agent workflows, and evaluation of prompts and retrieval. Provides technical direction, code review, and mentorship while staying close to implementation.

CORE TECHNOLOGIES

Languages and frameworks. Python, Java, Node.js, TypeScript, JavaScript, FastAPI, Django, Flask, React, Next.js, HTML, CSS, Tailwind CSS, SQL

AI and data. LLM APIs, RAG, LangChain, LangGraph, embeddings, pgvector, semantic retrieval, agent workflows, prompt evaluation, PostgreSQL, Prisma, Redis

Cloud and engineering. AWS, Docker, Kubernetes, Terraform, CI/CD, REST APIs, asynchronous processing, observability, production operations

Frontend. React, Next.js, TypeScript, JavaScript, HTML, CSS, Tailwind CSS, responsive UI, accessibility

PROFESSIONAL EXPERIENCE

Arbitexia | Engineering Manager, AI Platforms

Jan 2025 to Present Â· Remote from Toronto, Canada

â€¢ Provide hands-on technical leadership for Python-based AI applications, including service design, implementation, reviews, and delivery planning.
â€¢ Design FastAPI services and RAG workflows for document ingestion, semantic retrieval, and source-grounded responses.
â€¢ Prototype LLM agent workflows with tool integrations, explicit validation, audit trails, and human approval for consequential actions.
â€¢ Guide engineers on testing, prompt and retrieval evaluation, deployment practices, and operational troubleshooting.
â€¢ Collaborate with stakeholders to define feasible releases and balance product needs with maintainable system architecture.

Wealthsimple | Senior Software Engineer

Apr 2019 to Dec 2024 Â· Toronto, Ontario, Canada

- â€¢ Developed backend services and APIs for financial data workflows, integrations, and internal operational tools.
- â€¢ Designed asynchronous processing patterns with retries, idempotency, and observability to improve service resilience.
- â€¢ Optimized PostgreSQL queries and caching paths, and strengthened automated tests, release checks, and incident diagnostics.
- â€¢ Partnered with product, security, and infrastructure teams on system design and production rollouts, and mentored engineers through reviews and pairing.
- â€¢ Built internal React and TypeScript interfaces that surfaced workflow status and supported operational investigations.

Wave Financial | Software Engineer

Jul 2015 to Mar 2019 Â· Toronto, Ontario, Canada

- â€¢ Built and maintained Python and Django services and REST APIs for small-business financial application workflows.
- â€¢ Implemented background jobs, external API integrations, and data synchronization with clear failure handling.
- â€¢ Improved SQL access patterns, automated test coverage, and logging while collaborating with product and frontend teams.

Wave Financial | Junior Software Engineer

Jul 2012 to Jun 2015 Â· Toronto, Ontario, Canada

- â€¢ Delivered application features using Python, Django, JavaScript, and relational databases under senior engineering guidance.
- â€¢ Investigated defects, implemented validation and tests, and supported routine deployments and third-party integrations.

SELECTED ENGINEERING WORK

AI knowledge assistant. A RAG architecture combining ingestion, chunking, vector search, retrieval APIs, and grounded answers. Ingestion and chunking feed vector search, and retrieval APIs keep responses tied to the documents that were retrieved.

Workflow automation. A tool-calling system with deterministic checks and approval steps. Checks sit ahead of the action, and consequential actions require an explicit approval step.

EDUCATION

University of Waterloo | Bachelor of Computer Science

2008 to 2012 Â· Waterloo, Ontario

- â€¢ Finished a Bachelor of Computer Science, the academic base for a career in production software.
- â€¢ Built the grounding in software design, systems, and data used in the later backend and AI work.
- â€¢ Moved from Waterloo into professional engineering in Toronto right after graduation.